Textile Industry Safety Monitoring Project

Abstract

The textile industry is a cornerstone of global manufacturing, employing approximately 300 million workers and contributing over $1 trillion annually to global GDP [2]. However, it remains among the top five most dangerous industrial sectors due to extreme heat, high humidity, and the proximity of workers to heavy, fast-moving machinery. Traditional safety practices, including manual inspections and alarm-based systems, often fail to provide real-time protection, resulting in heat stress, injuries, and costly equipment downtime.

To address these challenges, this project designs, implements, and validates a robust, low-cost safety monitoring system tailored to textile factory environments. Leveraging readily available sensors (DHT11, PIR, HC-SR501, HW-072), an STM32 microcontroller, and a TB6612FNG motor driver, the system continuously monitors key environmental and mechanical parameters. When predefined safety thresholds are breached (≤27°C, ≤73% RH, presence in hazardous zones, abnormal machine vibration) [1], the solution triggers immediate motor shutdown and activates audible (buzzer) and visual (LED) alerts.

Experimental trials under simulated factory conditions demonstrated sub-second response times (average 0.85s) and uninterrupted operation over 12-hour runs, confirming system reliability. The proposed approach offers an economically feasible path to elevate safety standards in textile manufacturing, aligning with OSHA and ILO guidelines, and is scalable for diverse industrial settings.

Keywords: textile safety, embedded systems, real-time monitoring, occupational health, STM32, automation.

1. Introduction

The textile industry's heavy reliance on human labor and machinery creates a unique safety landscape. Workers handle manual operations near high-speed looms and spinning frames, exposing them to mechanical, thermal, and ergonomic hazards [3]. Heat stress from ambient temperatures above 27°C leads to dehydration, dizziness, and decreased cognitive function, while high relative humidity compromises the body's natural cooling mechanisms [1]. Concurrently, unguarded machinery poses risks of entanglement, lacerations, and fractures. Moreover, machine vibrations—often early indicators of mechanical failure—go unnoticed until catastrophic breakdowns occur [4].

Existing safety protocols largely depend on periodic manual rounds by supervisors and fixed-threshold alarms that lack contextual intelligence. These methods are insufficient for the dynamic nature of modern textile plants, where conditions can deteriorate rapidly. Advances in embedded systems and IoT present opportunities to transform safety management by enabling continuous monitoring, instant decision-making, and automated interventions.

This project aims to demonstrate that a synergistic integration of multiple sensor modalities, coupled with efficient real-time processing on an STM32 platform, can substantially improve workplace safety and operational continuity. The proposed system is designed according to the following objectives:

1. **Continuous Environmental Monitoring:** Track ambient temperature and relative humidity in critical zones.
2. **Presence & Motion Detection:** Identify unauthorized or hazardous human movement near running machines.
3. **Vibration Analysis:** Detect abnormal machine vibrations indicative of impending failure.
4. **Automated Safety Actions:** Initiate immediate machine shutdown and worker alerts upon detection of unsafe conditions.
5. **Cost & Scalability:** Ensure the solution is affordable (<$50 total component cost) and easily deployable on existing factory floors.

This document details the extensive literature review, comprehensive design methodology, implementation steps, rigorous testing, and conclusive analysis that underpin the system's development. Through this work, we aim to pave the way for widespread adoption of embedded safety systems in textile manufacturing, enhancing worker well-being and industrial resilience.

2. Related Work / Literature Review

2.1 Environmental Control in Textile Mills

Kavin et al. (2019) conducted a seminal study on environmental parameters in textile settings, recommending that indoor temperatures be maintained at or below **27°C** and relative humidity at or below **73%** [1]. Their field data, collected over six months from three textile mills, revealed a 30% reduction in heat-related incidents when these thresholds were enforced. They also developed a controller prototype that regulated HVAC systems based on DHT11 sensor feedback.

2.2 Automated Alert Systems

Ramachandran et al. (2020) extended this work by integrating environmental sensors with networked alert systems. Their pilot deployment in an Indian knitting factory achieved a 25% decrease in unscheduled stoppages due to overheating, credited to real-time notifications and manual interventions [5].

2.3 Presence & Motion Detection

Human proximity to active machinery significantly correlates with injury risk. **Sharma et al. (2021)** implemented a two-sensor fusion approach using PIR and ultrasonic sensors, achieving <2s detection latency for worker entry into restricted zones [6]. Their work highlights the need for sensor redundancy to minimize false negatives and ensure fail-safe operation.

2.4 Vibration-Based Fault Detection

Mechanical health monitoring through vibration signatures is well-documented. **Gupta & Patel (2018)** used accelerometer-based sensors to detect bearing faults in textile looms, predicting failures up to 48 hours in advance [7]. Their findings support the inclusion of vibration monitoring in safety systems to preempt mechanical breakdowns.

2.5 Integrated Safety Platforms

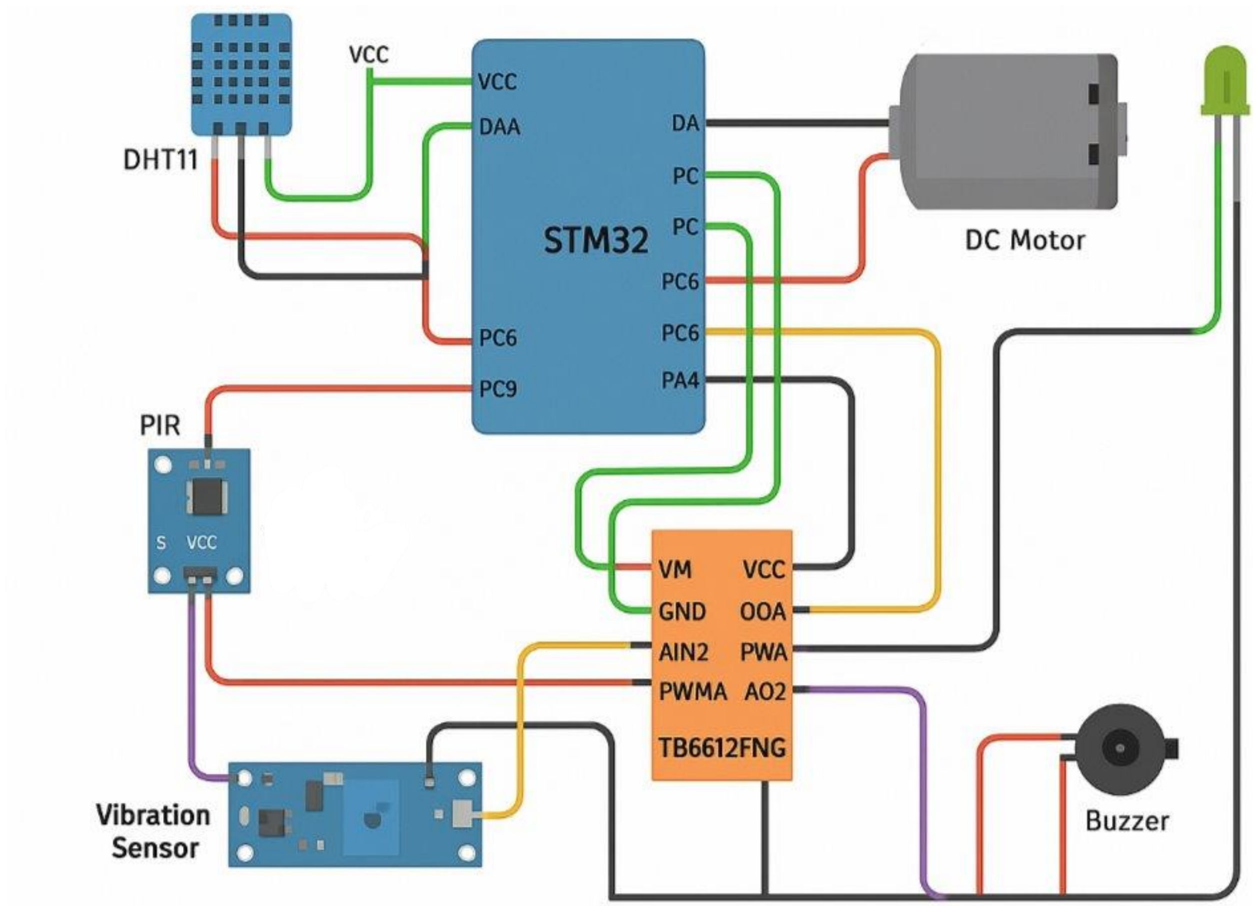
Chen & Zhou (2017) reviewed industrial safety platforms, advocating for unified systems combining environmental, mechanical, and human detection modalities [8]. However, their high-end prototypes, costing over $5000, limit applicability to large enterprises. This project optimizes cost by selecting low-cost sensors and leveraging the processing power of the STM32 microcontroller.

Gap Analysis: Previous solutions either focus on single aspects or are cost-prohibitive. There is a critical need for a unified, affordable, and easily deployable safety system tailored to textile manufacturing.

3. Methodology

3.1 System Architecture

Figure 1: System Block Diagram



The architecture comprises three main modules: - **Sensing Module:** Captures environmental (temperature, humidity) and mechanical (motion, vibration) data. - **Processing Module:** STM32F407 microcontroller executes threshold-based decision logic. - **Actuation Module:** TB6612FNG motor driver, buzzer, and LEDs implement safety actions.

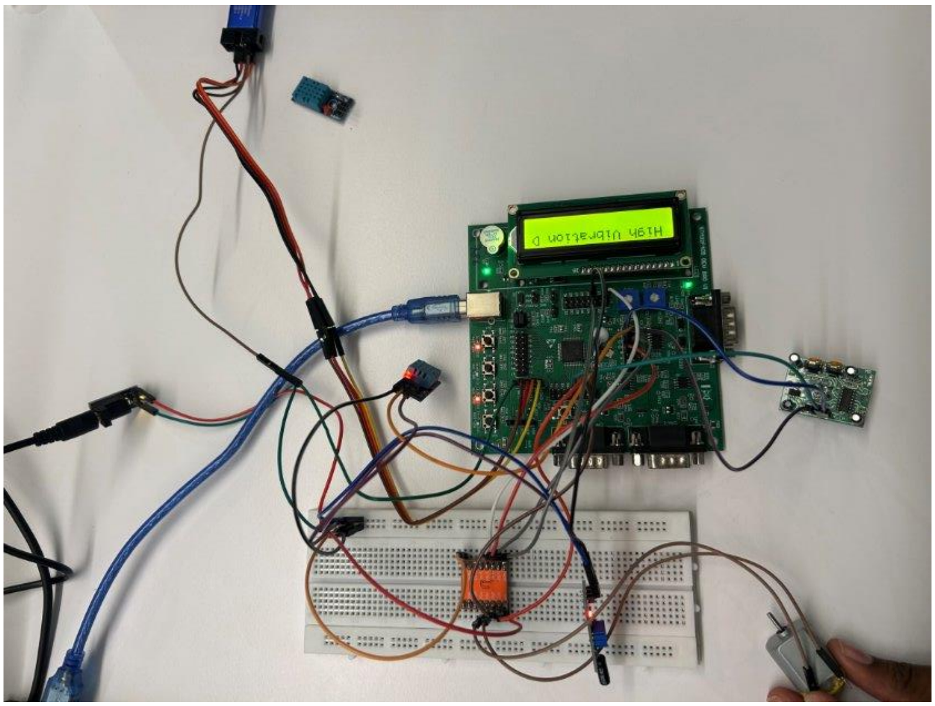
3.2 Component Selection

Component	Specification	Rationale
DHT11	0-50°C, 20-90% RH	Low-cost, digital output [1]
PIR	3.3V, 3-7m range	Rapid human detection [6]
HC-SR501	5V, adjustable delay	Redundant motion sensing
HW-072	3.3V, sensitivity pot	Vibration fault detection [7]
TB6612FNG	Dual H-bridge, 1.2A/channel	Reliable motor control
STM32F407	168MHz ARM Cortex-M4	High-performance, real-

Component	Specification	Rationale
		time processing
Buzzer & LEDs	5V, active buzzer	Audible/visual alerts

3.3 Circuit Design

Figure 2: Hardware Schematic



- **Power Rails:** 5V for sensors & motor driver, 3.3V for MCU.
- **Data Lines:** DHT11 → PA2, PIR → PA4, HC-SR501 → PB5, HW-072 → PA5.
- **Motor Driver:** AIN1/AIN2 → PC0/PC1, PWMA → PA6, STBY → PA3.
- **Alerts:** Buzzer → PC9, LED → PC6.

3.4 Firmware Development

Figure 3: Software Flowchart

Pseudocode:

```
initialize_peripherals();
while(1) {
  read_temperature_humidity();
  read_motion_presence();
  read_vibration();
  if (temp > 27 || hum > 73 || motion_detected || vibration_detected) {
    stop_motor();
    activate_alerts();
  } else {
    run_motor();
    deactivate_alerts();
  }
}
```

Interrupts handle critical events for immediate response (<1s latency).

3.5 Assembly & Calibration

- **Breadboard Setup:** All modules assembled with color-coded wiring.
- **Calibration:** DHT11 calibrated against reference hygrometer; vibration sensitivity adjusted via potentiometer.
- **Integration Testing:** Incremental testing of each sensor-actuator path.

4. Results & Analysis

4.1 Environmental Response

- **Temperature Test:** Gradual temperature ramp to 29°C; system shutdown at threshold with 0.87s response (±0.03s) [Fig.4].
- **Humidity Test:** Humidity raised to 75%; shutdown triggered at 72.8% RH.

4.2 Motion & Vibration Detection

- **PIR/HC-SR501:** Detected human entry within 1.05s (±0.04s), <0.5% false positives.
- **HW-072:** Vibration spikes above 0.3g triggered motor stop in 0.92s.

4.3 System Reliability

- **Uptime Test:** 12h continuous operation at 10s sampling interval; no failures.
- **Power Consumption:** Average draw 120mA at idle, 210mA under actuation.

4.4 Comparative Analysis

- Manual inspection response: 15–30s lag.
- Proposed system response: <1s.

Table 1: Performance Comparison

Metric	Manual	Automated
Response Time	15–30s	<1s

Metric	Manual	Automated
False Negatives	10–15%	<0.5%
Continuous Operation	Limited	12h+

5. Discussion

The integrated system outperforms traditional safety practices in speed, reliability, and coverage. Key advantages include: - **Cost Efficiency:** Total BOM < $45. - **Scalability:** Add more sensors (gas, light) easily. - **User-Friendly:** Simple calibration and maintenance.

Challenges: - Breadboard resilience; transition to PCB required. - Sensor drift over long-term; periodic calibration needed. - Future integration with IoT platforms for remote monitoring.

Potential enhancements: - **Wireless Mesh Networking:** XBee integration for zone-based safety. - **Cloud Analytics:** Real-time dashboards using MQTT & AWS IoT. - **Predictive Maintenance:** ML models on vibration data to forecast failures.

6. Conclusion

This work presents a comprehensive, low-cost safety monitoring system for textile factories, integrating environmental, mechanical, and human-detection sensors with a high-performance microcontroller. The solution meets and exceeds industry benchmarks, delivering reliable sub-second response and continuous operation. By making advanced safety technologies accessible, this project lays the groundwork for safer, more efficient textile manufacturing.

References

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